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Original article

Implementation of intensity-modulated radiotherapy for head and neck cancers in routine practice☆

Évaluation de la radiothérapie conformationnelle avec modulation d'intensité utilisée en pratique quotidienne pour les carcinomes de la tête et du cou

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A B S T R A C T

Purpose. – To report on patterns of relapse following implementation of intensity-modulated radiotherapy and subsequent changes in practice in a tertiary care centre.
Patients and methods. – Between 2008 and 2011, 188 consecutive patients (mean age 59 years old) received intensity-modulated radiotherapies with curative intent for squamous cell carcinomas of the oral cavity (17.5%), oropharynx (43%), hypopharynx (21%), larynx (14%), sinonasal cavities (6%), nasopharynx (1.5%) at the university hospital of Besançon. There were stage I and II 9%, III 24.5%, IV 66.5%. One hundred and thirty-eight underwent exclusive intensity-modulated radiotherapy, 50 underwent postoperative intensity-modulated radiotherapy, 174 had concurrent chemotherapy, 57 had induction chemotherapy. Dynamic intensity-modulated radiotherapy with static fields was performed for all patients using sequential irradiation in 174 patients and simultaneous integrated boost irradiation in 14 patients.
Results. – With a median follow-up was 27.5 months, there was 79% of locoregional failures occurred in the 95% isodose. Two-year overall survival, disease-free, local failure-free and locoregional failure-free survival rates were73%, 60%, 79% and 72%, respectively. Prognostic factors for disease-free survival were stage (IV vs. I–III) with a relative risk of 1.7 [1.1–2.8] ($P = 0.02$) and T stage with 1.6 [1.04–2.5] ($P = 0.03$).
Conclusion. – The current series showed similar patterns of failure as in other tertiary care centres. We did not identify intensity-modulated radiotherapy specific relapse risks.

Mots clés :
Planification de traitement
RCMI
Carcinomes de la tête et du cou
Échecs locorégionaux

R É S U M É

Objectif de l'étude. – Identifier les sites de récidives locorégionales dans les carcinomes de la tête et du cou après radiothérapie conformationnelle avec modulation d'intensité (RCMI) utilisée en pratique quotidienne.
Patients et méthodes. – Entre 2008 et 2011, 188 patients consécutifs âgés en moyenne de 59 ans (27–82), atteints d'un carcinome de la cavité buccale (17,5 %), de l'oropharynx (43 %), de l'hypopharynx (21 %), du larynx (14 %), du sinus maxillaire (6 %) et du nasopharynx (1,5 %) ont reçu une RCMI avec intention curative au centre hospitalier régional universitaire de Besançon. La répartition par stade était I et II

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